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Full name of presenter / Nombre y apellidos del presentador

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Title / Título

Prediction of CYP450 Metabolizer Phenotypes from SNP Array Data Using GenoStaR: A Pharmacogenetic study in the MadManic cohort

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Abstract (max. 300 words) / Resumen (máx. 300 palabras)

Background: Pharmacogenetics plays a key role in personalized medicine by linking genetic variation to differences in drug response. Variants in cytochrome P450 genes influence both drug efficacy and adverse effects. Since CYP450 enzymes metabolize 70-80% of clinically used drugs, accurate genotype-to-phenotype translation is essential for optimizing pharmacological therapies. In this context, GenoStaR is an R package that converts genotypes into star alleles and predicts metabolic phenotypes for major CYP450 genes using SNPs, insertion/deletion variants, and structural variants.

Objective: Validate GenoStaR in 660 genotyped individuals from the MadManic cohort to obtain diplotypes, activity scores, and metabolic phenotypes for the main CYP450 genes (CYP1A2, CYP2B6, CYP2C9, CYP2C19, CYP2D6, CYP3A4 and CYP3A5).

Methods: SNVs and SNPs located within the seven analyzed CYP450 genes were extracted with PLINK, resulting in a total of 1260 variants. GenoStaR was subsequently used to infer the associated metabolic phenotype from genotype data through `all_genotype_pheno()` function, using the option `phased = FALSE` because the data were unphased, and the “new” star allele nomenclature based on the PharmVar December 2024 update for CYP1A2. Internally, `all_genotype_pheno()` uses `assign_diplotype()` to infer CYP450 diplotypes from the observed genotypes. From these diplotypes, activity scores for CYP2C9 and CYP2D6 are calculated using `star_to_activity()`, while metabolic phenotypes for CYP2B6, CYP2C9, CYP2C19, CYP2D6 and CYP3A5 are inferred using `star_to_pheno()`.

Results: The error associated with variants not genotyped in our cohort but required by GenoStaR for the European population was very low, suggesting adequate coverage of pharmacogenetically relevant variants included in the analysis.

Call rates above 95% were obtained for CYP1A2, CYP2B6, CYP2C9, CYP3A4, CYP2C19 and CYP3A5, demonstrating a high capacity for diplotype inference. Moreover, comparison of star allele frequencies, activity scores, and metabolic phenotypes with ClinPGx European population frequencies showed high concordance. In contrast, CYP2D6 showed lower call rate (47.6%) and greater discrepancies with ClinPGx, likely due to limitations in the assignment of complex diplotypes in these gene.

Discussion: GenoStaR represents a fast and useful tool for pharmacogenetic prediction of metabolic phenotypes from genotyping data, although further improvements are required to increase call rate and accuracy for CYP2D6.